An explicit initial Θ-datum with mixed reduction at a split prime

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Abstract

We exhibit an explicit initial Θ-datum in the sense of [IUT1, Def. 3.1]: an elliptic curve in Legendre form over $\mathbb{Q}(\sqrt{5})$ with $\ell = 7$, for which the rational prime 211 *splits* in the field of moduli, so that one place of multiplicative and one place of good reduction lie above it. Every condition (a)–(f) of [IUT1, Def. 3.1] is verified individually, several by exact computation, and the local invariants are $e_{\mathrm{g}} = 1$, $e_{\mathrm{b}} = 105$ and $\mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = 29/7$, the profile being tame at 211 since $105 \leq 209$.

The configuration is not incidental. Because [IUT1, Def. 3.1(e)] makes $\mathbb{V}$ a *section* of $\mathbb{V}(K) \twoheadrightarrow \mathbb{V}_{\mathrm{mod}}$, the tuples indexing a local tensor packet at a rational prime are constant unless that prime has several places in $\mathbb{V}_{\mathrm{mod}}$; and for the curve constructed here the only odd non-split places of multiplicative reduction are 3 and 5, carrying less than 1.24% of the relevant Tate contribution outside 2ℓ. Nor can the configuration be removed by a different choice of $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$: the additional good-reduction hypothesis of [IUT4, Thm. 1.10] does not permit an actual bad place outside 2ℓ to be treated as chosen-good.

Under one hypothesis (SA) about the source, isolated in §5 and discussed in §7, the upper bound obtained in [IUT4, Thm. 1.10, proof, Step (v)] by applying [IUT4, Prop. 1.2(iii)] to a fixed target component is incompatible with the region it is applied to, with margin $8/7$ at $j = 1$ and $88/7$ at $j = 2$. We state (SA) separately, list the ways it can fail, and undertake to withdraw that application if the reading is shown to be wrong; we would regard such a demonstration as the more valuable outcome, since it would make explicit what the collection of possible images is at a tuple of mixed reduction type.

We ask how the admitted configuration is treated (Question 1). All arithmetic is exact and is reproduced by a short script.

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1 Introduction

Explicit initial Θ-data are needed for any numerically effective use of the theory; [MFHMP] is devoted to obtaining such uses. This note contributes one, constructed so that every condition of [IUT1, Def. 3.1] can be checked individually, and then asks how a configuration it exhibits is handled.

The datum is stated in §3 and verified condition by condition. §4 shows that the configuration it exhibits is not incidental, in two senses: it is forced by the shape of [IUT1, Def. 3.1(e)] (Proposition 2), and for this curve it cannot be removed by choosing $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ differently (Proposition 4, Remark 3). §§5–6 draw a consequence, conditional on one hypothesis about the source which we isolate and state separately; §§7–8 say what would falsify it and what we ask. §9 records the computations.

Our purpose is to have the question answered. We do not argue that the theory fails; we were unable to locate the treatment of this configuration, and we would regard being shown it as the useful outcome. Two features of the situation should be said at once.

- The configuration cannot arise when $F_{\mathrm{mod}} = \mathbb{Q}$ (Corollary 3(i)). This mechanism alone consequently says nothing about estimates for such data. We draw no conclusion about the effective Diophantine results of [MFHMP].
- That the construction below is possible at all is a property of [IUT1, Def. 3.1]: its conditions (a)–(f) are stated precisely enough that each can be verified individually against an explicit curve, several of them by exact machine computation. Not every foundational definition admits this.

2 What is and is not claimed

We mark source statements [S], the present deductions and exact calculations [P], and the source-adapter reading [H]. These are different statuses: the arithmetic does not certify (SA). Statement locators for IUT I and III refer to the author's May 2020 PDFs, IUT II to the December 2020 PDF, and IUT IV to the April 2020 PDF; the bibliography records their 2021 publication metadata separately. For [SS] we use the manuscript dated July 16, 2018, and for [Rpt] the February 2019, 45-page version of the report on the March 2018 discussions.

Claimed. (i) The datum of §3 is an initial Θ-datum with a mixed local profile above 211. This is independent of (SA); the arithmetic in it is machine-checkable, and the remaining ingredients are standard facts and cited constructions, identified as such in §3. (ii) *Assuming* (SA), the local comparison of §6 fails, with margin $8/7$.

Not claimed. That the *abc* conjecture is false; that the lower bound of [IUT3, Cor. 3.12] is false; that the global inequality of [IUT4, Thm. 1.10] is false; that the objections of [SS] are correct or the responses of [Rpt] incorrect; that the estimate cannot be repaired. What is at issue is one application of one local bound.

3 The datum

$$\begin{aligned}
&F_0 = \mathbb{Q}(\sqrt{5}), && \pi = 16 + 3\sqrt{5}, \quad \mathrm{N}(\pi) = 211, \quad a = \pi^{29}, \\
&E_0 : y^2 = x(x-1)(x-a), && \ell = 7, \\
&F = F_0(i, E_0[15]), \quad i^2 = -1, && K = F(E_0[7]) = F_0(i, E_0[105]).
\end{aligned} \tag{1}$$

Let $X_F = E_F \setminus \{0\}$. Select as $\mathbb{V}^{\mathrm{bad}}_{\mathrm{mod}}$ all places of F_0 of multiplicative reduction outside $2, 7$, including the place v_π with $v_\pi(\pi) = 1$. The compatible section $\mathbb{V}$ and cusp ϵ are constructed below; they are not arbitrary choices.

Theorem 1. [P] *These data extend to an initial Θ-datum in the sense of [IUT1, Def. 3.1] and to an LGP-Gaussian log-theta-lattice. They satisfy the additional arithmetic hypotheses of [IUT4, Thm. 1.10]. Above* 211*,* $\mathbb{V}$ *has exactly two elements, one selected bad and one good, and*

$$e_{\mathrm{g}} = 1, \qquad e_{\mathrm{b}} = 105, \qquad \mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}) = \frac{29}{7}.$$

In particular every local index in this profile satisfies $e \leq p - 2 = 209$*.*

Here $\mathrm{ord}_p(p) = 1$; an integral prime-ideal order is denoted $v_{\mathfrak{p}}$. The following proof separates exact arithmetic from the geometric constructions. A detailed expansion is in [Full, §§3–5, especially §4.6].

Conditions (a) and (b). The field F contains i and full 6-torsion, the 2-torsion already being rational from the Legendre equation. Its degree over F_0 divides

$$2|\mathrm{GL}_2(\mathbb{Z}/15\mathbb{Z})| = 2 \cdot 48 \cdot 480 = 46080 = 2^{10}3^2 5,$$

and it is Galois. At every finite place, at least one of $3, 5$ is prime to the residue characteristic. Full rational torsion of that order gives semistable reduction by [IUT4, Prop. 1.8(v)]. The origin supplies the marked section; the resulting one-pointed genus-one stable model gives the required reduction of X_F.

The six Legendre orbit comparisons between a and $\bar{a}$ all fail, so $j(E_0) \neq \overline{j(E_0)}$ and $F_{\mathrm{mod}} = \mathbb{Q}(j(E_0)) = F_0$. At every odd place where a or $1 - a$ has positive order, the integral Legendre invariant $c_4 = 16(a^2 - a + 1)$ is a unit, proving multiplicative reduction directly. If $a \equiv 0$, the nodal tangent equation is $y^2 = -x^2$, which splits after adjoining i. If $a \equiv 1$, put $u = x - 1$; the nodal tangent equation is $y^2 = u^2$, already split. Thus every selected place is split multiplicative over F, since $i \in F$. At every remaining place outside $2, 7$ the discriminant $16a^2(1-a)^2$ is a unit. Consequently the selected bad set is the full one required for the extra good-reduction hypothesis in IV. Also $F_{\mathrm{tpd}} = F_0$ and $F = F_0(i, E_0[30])$ exactly, because $E_0[2]$ is already rational. This is the exact field in IV, not an arbitrary torsion-containing extension.

Condition (c), including every selected order. At the degree-one place 11, $\sqrt{5} = 4$, the reduction has trace zero; its characteristic polynomial modulo 7 has discriminant 5, a nonsquare. Hence the representation is irreducible. Tate inertia at v_π, where $v_\pi(q) = 58$, contains a nonidentity transvection on $E_0[7]$. Irreducibility provides a conjugate transvection with a different fixed line. Their powers generate the two opposite root groups, hence $\mathrm{SL}_2(\mathbb{F}_7)$. This group remains in the image over F: the latter image is normal of index prime to 7. Every proper normal subgroup of $\mathrm{SL}_2(\mathbb{F}_7)$ lies in its center, and the corresponding quotient has order divisible by 7. Intersecting with the latter image therefore cannot give a proper subgroup. Thus $\mathrm{Gal}(K/F)$ has the required image.

The exact norm certificate gives

$$\mathrm{N}(a) = 211^{29}, \qquad \mathrm{N}(1-a) = 2^2 3^2 \cdot 5 \cdot 59 \cdot 80621 \cdot C,$$

where

$$C = 29622281599338016545834674091843816951690863770835007573979.$$

The second norm is seventh-power-free, as certified by the exhaustive sieve of §9; primality of C is not asserted. The norm identity $v_q \mathrm{N}(z) = \sum_{\mathfrak{p}|q} f(\mathfrak{p}/q) v_{\mathfrak{p}}(z)$ shows that every positive order of $1 - a$ is at most 6. The only positive order of a is 29 at v_π. Write r for the relevant positive order at a selected place. Its base Tate order is $2r$, prime to 7. Over a place w of F it is $2re(w/\mathfrak{p})$, still prime to 7 because the ramification index divides $[F : F_0]$. No selected residue characteristic is 7. This verifies the all-place clause, not just the order 58 at v_π.

Condition (d): quotient and core. Fix one global line $W \subset V = E_0[7](K)$ and a nonzero $\xi \in Q = V/W$. Let $\pi_W : E_K \to E' = E_K/W$ and let $\psi : E' \to E_K$ satisfy $\psi\pi_W = [7]$. Then $\ker\psi \simeq Q$ is constant. Put

$$\underline{X}_K = E' \setminus \ker\psi, \qquad \underline{C}_K = [\underline{X}_K/\{\pm 1\}], \qquad C_K = [X_K/\{\pm 1\}].$$

The finite etale covering $\underline{C}_K \to C_K$ has type $(1, 7\text{-tors})^{\pm}$ by [EtTh, Def. 2.1]: its geometric quotient is onto Q, with zero decomposition-group action at the rational zero cusp. Let ϵ be the cusp corresponding to $\pm\xi$.

The four arithmetic exceptional j-invariants are

$$2^{14}31^3 5^{-3}, \quad 2^2 73^3 3^{-4}, \quad 1728, \quad 0 \qquad \text{[MFHMP, Prop. 2.1]}.$$

All are rational and hence exclude this j. The non-arithmetic punctured hemi-elliptic orbicurve $C_{\overline{K}}$ is itself a core by [CanLift, Prop. 2.7]. Proposition 2.3(i) there descends it to C_K, the core of the finite etale cover $\underline{C}_K$. For each completion, Proposition 2.3(ii) followed by (i) gives the specified core C_{K_v} as well. Non-CM or nonintegral j alone is not being used as the criterion. The geometric theta covers required in (d) are supplied by [EtTh, Prop. 2.2, Def. 2.3].

Conditions (e) and (f): compatible places and cusp. At an initially chosen lift of each selected moduli place, the split Tate sequence $0 \to \mu_7 \to E[7] \to \mathbb{Z}/7\mathbb{Z} \to 0$, with the class of $q^{1/7}$ mapping to 1, specifies a line and a nonzero quotient class. The group $\mathrm{SL}_2(V)$ is transitive on such marked flags: complete a lift of the quotient class to a determinant-one basis. Choose $\sigma \in \mathrm{Gal}(K/F)$ taking this local flag to the fixed $(W, \pm\xi)$, and replace the chosen place by its image under σ. The group acts on the places; the global quotient and cusp stay fixed. Do this separately over every selected moduli place, then choose lifts over the remaining places. This gives a section $\mathbb{V} \to \mathbb{V}_{\mathrm{mod}}$; equivariance of the section is not required.

Locally the quotient is $E(q) \to E(q^7)$, $z \mapsto z^7$, and its dual is $E(q^7) \to E(q)$, $z \mapsto z$. Thus the local cover has type $(1, \mathbb{Z}/7\mathbb{Z})^{\pm}$ and ϵ is its canonical ± 1 cusp. Full 2-torsion gives the field condition in [IUT1, Def. 3.1(e)]; the Weil pairings give $\mu_{12} \subset F$, $\mu_7 \subset K$. Together with the local core, these supply the arithmetic theta models of [EtTh, Thm. 1.10(iii), Def. 2.5(i)]. The arrow-marked covers and their open subgroups are then those of [IUT1, §1, Def. 1.1, Rem. 1.1.2], using the decomposition group of the cusp 2ϵ; $2\xi \neq \pm\xi$. The torsion assumption is on the original E_K, not full $E'[7]$. Their base changes give the covers at good places. This supplies the covering and cusp clauses as well as the section.

Lattice and IV input. The native models and bridge identifications in [IUT1, Prop. 4.8(ii), Prop. 6.7, Def. 6.13] give a full theater. On tagged copies, the synchronized full log-links of [IUT3, Prop. 1.3], the nonempty enhanced Gaussian isomorphisms of [IUT2, Cor. 4.10(iii)], and the incoming-link Gaussian/LGP identifications of [IUT3, Prop. 3.7] give the lattice of [IUT3, Def. 3.8(ii),(iii)]; see the construction in [Full, §4.6]. This existence uses neither (SA) nor a numerical consequence of [IUT3, Cor. 3.12]. The exact field and extra good-reduction conditions of IV were checked above.

Remark 1 (the local invariants)**.** At 211, the two base places have $\sqrt{5} = 65, 146$ modulo 211. The corresponding residues of a are $0, 26$; the first has order 29 and the second is good. Good reduction and prime-to-211 torsion give $e_{\mathrm{g}} = 1$. At the bad base place the nodal tangent directions are defined by -1, a nonsquare in $\mathbb{F}_{211}$. Over the unramified quadratic field $k_0 = \mathbb{Q}_{211}(i)$ the curve is split Tate with $\mathrm{ord}_{211}(q) = 58$. For $n = 15, 105$, its exact torsion field is $k_0(E[n]) = k_0(\zeta_n, q^{1/n})$.

The denominator of $58/n$ forces its ramification index to be a multiple of n; [IUT4, Prop. 1.8(vii)] bounds that index by a divisor of n. It is therefore exactly n. The chosen completions of F and K have indices 15 and 105. Finally the 2ℓ-th root satisfies $\mathrm{ord}_{211}(\underline{\underline{q}}) = 58/14 = 29/7$. No claim that a general root adjunction is totally ramified is needed.

4 When can this configuration occur at all?

The following is independent of everything below; it uses only [IUT1, Def. 3.1(e)] and [IUT3, Prop. 3.1].

Proposition 2. *Let p be a rational prime and $\mathbb{V}_p = \{v \in \mathbb{V} : v \mid p\}$. Then $|\mathbb{V}_p|$ equals the number of places of F_{mod} above p. Consequently, in the decomposition $\log({}^{A}F_{v_{\mathbb{Q}}}) = \bigoplus_{\{v^\alpha\}_{\alpha\in A}} \bigotimes_{\alpha \in A} \log({}^{\alpha}F_{v^\alpha})$ of [IUT3, Prop. 3.1], the tuples $\{v^\alpha\}$ at p are all constant unless p has at least two places in $\mathbb{V}_{\mathrm{mod}}$.*

Proof. [IUT1, Def. 3.1(e)] makes $\mathbb{V}$ a section of $\mathbb{V}(K) \twoheadrightarrow \mathbb{V}_{\mathrm{mod}}$, and a section restricts to a bijection above each rational prime. The tuple index set is $\mathbb{V}_p^A$. □

Corollary 3. [P] *At a stage of length $k = j+1 \geq 2$, write $r = |\mathbb{V}_p|$. There are $r^k - r$ nonconstant place tuples, so they occur exactly when $r \geq 2$, independently of reduction type. If the chosen-bad and chosen-good subsets have sizes b, g, mixed-status tuples number $r^k - b^k - g^k$ and occur exactly when $b, g > 0$. In particular:*

- *(i) if $F_{\mathrm{mod}} = \mathbb{Q}$, every tuple at every prime is constant;*
- *(ii) in a quadratic moduli field, a nonsplit rational prime has a single place, hence only a constant tuple. A split prime has nonconstant tuples, and has mixed-status tuples precisely when one place is chosen bad and the other chosen good.*

Remark 2 (rational moduli data)**.** By Corollary 3(i), the configuration examined in this note *cannot* arise when $F_{\mathrm{mod}} = \mathbb{Q}$. This excludes the present mixed-place mechanism for such data. It does not identify every datum used in an effective Diophantine argument with a rational-moduli datum, and we make no claim about the validity of the effective results in [MFHMP].

Proposition 4 (the configuration cannot be removed by a choice)**.** *Write $a = \pi^{29} = A + B\sqrt{5}$ with $A, B \in \mathbb{Z}$. Then $\gcd(A, B) = 1$ and $\gcd(A-1, B) = 3$, and consequently the only odd rational primes of multiplicative reduction that do* not *split in F_0 are 3 and 5.*

Proof. Let p be an odd prime inert in F_0, so $\sqrt{5} \notin \mathbb{F}_p$. Then $a \equiv 0$ would force $A \equiv B \equiv 0$, i.e. $p \mid \gcd(A, B) = 1$, which is impossible; and $a \equiv 1$ would force $A - 1 \equiv B \equiv 0$, i.e. $p \mid \gcd(A-1, B) = 3$, so $p = 3$ — which is indeed inert, 5 being a non-residue modulo 3. The remaining odd non-split prime is the ramified $p = 5$. □

Remark 3 (what a restriction to non-split places would cost)**.** Write H_{all} and H_{ns} for the degree-normalized Tate contributions outside 2ℓ from all selected bad places and from the non-split ones respectively. Then $H_{\mathrm{all}} = 29 \log 211 + \log\bigl(\mathrm{N}(1-a)/4\bigr)$, $H_{\mathrm{ns}} = \log 45$, and $H_{\mathrm{ns}}/H_{\mathrm{all}} < 0.0124$, so such a restriction would discard more than 98.76% of that quantity. (This is not a ratio of full Weil heights.) Independently of the cost, the additional good-reduction hypothesis imposed in [IUT4, Thm. 1.10] does not permit an actual bad place outside 2ℓ to be treated as chosen-good, so the substitution is unavailable in any case.

Question 1. [S] Definition 3.1(b) of [IUT1] permits a nonempty subset of the multiplicative places of odd residue characteristic and does not impose a splitting restriction. Its chosen-good places may still have bad reduction. The additional good-reduction hypothesis of [IUT4, Thm. 1.10] removes that freedom outside 2ℓ: every actual bad place there must be selected. Thus omitting v_π from the present datum would prevent that IV application.

For this admitted mixed profile, how does the fixed-tuple estimate in Step (v) apply to the possible-image hull after the specified factor transports? If a further restriction on the data is required, which stated hypothesis imposes it, and which applications remain within its scope?

Remark 4 (related work, added after the first deposit)**.** We learned of the following only after depositing the first version of this note.

[S] [LANA] isolates a compatibility problem in the passage from [IUT3, Thm. 3.11] to [IUT3, Cor. 3.12]. Its main goal [LANA, §9.2] is to show that $\eta^q = \eta_S^{\mathrm{anab}}$ for some suitable S, where the two sides are isomorphisms of pointed real lines obtained respectively from the q-pilot in its native structure and by anabelian and Kummer-theoretic procedures; [LANA, §9.3] reads this as the assertion that the rigidified q-pilot is represented in the output regions. [LANA] states that it does not have a proof of this at present, and reserves judgement.

[P] That is not the same statement as (SA)4. (SA)4 concerns which regions the union of possible images contains, and (F2) concerns the reading of the resulting hull in the fixed target of Step (v) of [IUT4, Thm. 1.10]. Within [LANA] the corresponding place is [LANA, §10.4], which locates the effect of the indeterminacies in the ambiguity of the choice of S and says that this is reflected in the log-volume computation for the hull in [IUT4].

[S] [LANA, §10.4] continues that the hull computation "is made at each place individually, but the effects of (Ind1), (Ind2) and (Ind3) cannot be detected locally". [P] This does not contradict Theorem 5, which is conditional: if that reading is correct then (SA) fails, and what is withdrawn is the source application, not the rational calculation. Nor have we identified the regions, admitted transports, hull and normalized readout of [LANA] with the fixed mixed tensor component used here; without that identification the two statements are not yet about the same objects. We record the cross-check as (F5) in §7.

[S] One point of contact is exact. The LGP element of [LANA, §5.2(f)] is $\left(\bigotimes_{t\in S_j}(1)_{w|v_{\mathbb{Q}}}\right)\otimes \underline{\underline{q}}_v^{\,j^2}$ at a bad place and $\left(\bigotimes_{t\in S_j}(1)_{w|v_{\mathbb{Q}}}\right)\otimes 1$ at a good one: the factor $\underline{\underline{q}}_v^{\,j^2}$ occupies the distinguished slot alone and every other slot carries 1. That is the same shape as the reading of Step (v) used in §6, and as the insert-1 representation of §5.

[S] [LANA] gives no explicit numerical initial Θ-datum [LANA, §2], and computes no fibre cardinality and no separate constant/non-constant/mixed classification of tuples. Its containers are nevertheless indexed exactly as here: [LANA, §5.2(a)] sets $K_{S_{j+1},v} = \left(\bigotimes_{t\in S_j}\bigoplus_{w|v_{\mathbb{Q}}}K_{t,w}\right)\otimes K_{j,v}$, so the decomposition of Proposition 2 is present in that formalism and the configuration of §4 is admitted by it. What §4 adds is that the configuration occurs, and §3 that it occurs in an admitted datum.

[S] [Form] decomposes the same passage as 3.11 $\Rightarrow$ 3.11.5 $\Rightarrow$ 3.12, obtained by moving the "hull+det" of 3.12 into 3.11.5. We use this only for the decomposition. We do not identify the transport of (SA)3 with the algorithmic parallel transport of [IUT3, Rem. 3.11.1(iv)]. That remark says of the parallel transport *mechanism* that it "does not consist of a simple instance of transport of some set-theoretic region ... via some set-theoretic function", but is a construction algorithm valid in the arithmetic holomorphic structures on both sides of the Θ-link. The map $\tau_{\sigma,t}$ of §5 is of a different type: it is a map between summands of one fixed ambient E on one side. We do not read the remark as a prohibition on such a component map, and we do not borrow the name for it. The label "3.11.5" is explanatory; [Form] states that it does not appear in [IUT1, IUT2, IUT3, IUT4].

The selection and tuple statements preceding this question are independent of (SA). Our proposed interpretation of the comparison itself is conditional on (SA), as specified next. We do not infer an independent probability for the splitting of bad primes from a density theorem.

5 The hypothesis (SA)

Fix one lattice of Theorem 1, a column $(n,\circ)$, a stage j and $p = 211$. Write $E = I^{\mathbb{Q}}(S^{\pm}_{j+1}; {}^{n,\circ}D^{\vdash}_{p})$ for the ambient, $U^{\mathrm{act}}_{j,t}$ for the t-component of the union of possible images of a Θ-pilot object in the sense of [IUT3, Cor. 3.12], $C_{\widetilde{\mathcal{O}}_t}$ for the terminal holomorphic hull, and $L_t = \bigotimes_i \log \mathcal{O}^{\times}_{K_{t_i}}$. The source writes the tuple decomposition explicitly: [IUT3, Prop. 3.1] gives $\log({}^{A}F_{v_{\mathbb{Q}}}) = \bigotimes_{\alpha\in A} \log({}^{\alpha}F_{v_{\mathbb{Q}}}) = \bigoplus_{\{v^{\alpha}\}_{\alpha\in A}} \bigotimes_{\alpha\in A} \log({}^{\alpha}F_{v^{\alpha}})$, "the direct sum ... over all collections $\{v^{\alpha}\}_{\alpha\in A}$", and [IUT3, Prop. 3.2] the mono-analytic analogue. With $A = S^{\pm}_{j+1}$ the summands are indexed by tuples $t \in \{\mathrm{g},\mathrm{b}\}^{j+1}$; (Ind1) permutes the outer label factors, while (Ind2) acts within the place-indexed factors and does not freely permute the places. Writing $\mathrm{Peel}_{\alpha,v} = \bigoplus_{t\,:\,t_{\alpha}=v} V_t$, a whole-factor permutation satisfies $T_{\sigma}(\mathrm{Peel}_{\alpha,v}) = \mathrm{Peel}_{\sigma(\alpha),v}$: *the place is unchanged and only the position moves.* In particular a coefficient is never re-created at the target slot.

Hypothesis 1. [H] In the fixed target arithmetic structure and normalization,

$$\begin{cases} L_t \subseteq C_{\widetilde{\mathcal{O}}_t}\big(U^{\mathrm{act}}_{j,t}\big), & t \text{ has at least one good factor}, \\ p^{\lceil j^2N/7\rceil} L_t \subseteq C_{\widetilde{\mathcal{O}}_t}\big(U^{\mathrm{act}}_{j,t}\big), & t = (\mathrm{b},\dots,\mathrm{b}), \end{cases} \qquad N = 7\,\mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}).$$

(SA) is a *reading* of the source, not a computation. Its proposed derivation has four steps. The cited constructions are [S]; their identification with the admitted regions and the comparison target in these steps remains [H].

- (SA1) *One output exists.* [IUT3, Prop. 3.7(ii),(v)] makes J_v "a subset [indeed a submodule when $v \in \mathbb{V}^{\mathrm{non}}$]"; [IUT3, Def. 3.8(i)] defines the Θ-pilot from generators *indexed by* $v \in \mathbb{V}^{\mathrm{bad}}$ *only*; [IUT3, Prop. 3.10(i)] gives native / coric compatibility; [IUT3, Thm. 3.11(ii)(a)] gives isomorphisms of the tensor packets *and* of their $\mathbb{Q}$-spans, compatible with log-volumes. Our reading realizes the specified native representation as one region $\kappa_E[J]$ among the possible images. It does not admit every isomorphic fractional ideal or replace J by its full span.
- (SA2) *Which automorphisms are admitted.* In [IUT1, §0, Def. 4.10] the arrows of a procession are sets of capsule-full poly-morphisms, so a factor transposition is realized by one automorphism of the whole procession.
- (SA3) *The type of the action.* [IUT3, Thm. 3.11(i)] treats the data "regarded up to indeterminacies", (Ind1) being "the indeterminacies induced by the automorphisms of the procession", and states the algorithm is "functorial with respect to isomorphisms of processions". Labels, coefficients and realizations are transported together.
- (SA4) *Same comparison target.* [IUT3, Cor. 3.12] forms the *union* of the possible images "which we regard as subject to the indeterminacies (Ind1), (Ind2), (Ind3)", and only then the hull. Our reading includes the whole transported representation in this same union, including its global object and localizations, so that its $\mathcal{F}^{\Vdash}$-prime-strip restriction is preserved. It also identifies the resulting fixed arithmetic target and normalized readout with those used in Step (v) of [IUT4, Thm. 1.10].

Under these four clauses, choose $s = \sigma^{-1}t$ with a good factor in its distinguished slot. The native ideal generated by the insert-1 representation is $J_s = \widetilde{O}_s$ by [IUT3, Prop. 3.1(ii), Prop. 3.4, Example 3.6(ii), Prop. 3.7(v), Def. 3.8(i)]. Here $\widetilde{O}_s$ is the integral closure of the tensor order $R_s = \bigotimes_i \mathcal{O}_{K_{s_i}}$, and $L_s \subseteq R_s \subseteq J_s$ because every $e_i \leq p-2$. The ind-topological additive comparison in [IUT3, Prop. 3.2, Thm. 3.11(ii)(a)] preserves the integral tensor packet I_s and its rational span. On these finite-dimensional portions, continuity gives $\mathbb{Q}_p$-linearity. Since $L_s = p^{j+1} I_s$ by [IUT3, Rem. 1.2.2(i)], it also preserves L. Transporting the whole representation, taking the $\mathbb{Z}_p$-span and then the terminal target hull gives the first inclusion of (SA). For the all-bad tuple, put $c = \lceil j^2 N/7 \rceil$ and let a_s denote the generator, at the distinguished slot of s, of the native local fractional ideal J_s of [IUT3, Prop. 3.7(v)] — by [IUT3, Def. 3.8(i)] a generator up to torsion of the monoid indexed by that place, so $\mathrm{ord}(a_s) = \mathrm{ord}(\underline{\underline{q}}^{j^2}) = j^2 N/7$. Then p^c/a_s is integral and $p^c L_s \subseteq p^c R_s \subseteq a_s R_s \subseteq J_s$. Integer-scalar linearity gives the second inclusion. No ring linearity of κ_E, or commutation of that map with the hull, is asserted.

In particular the transport has the typed component equation

$$\mathrm{pr}_t\, T_\sigma = \tau_{\sigma,t}\, \mathrm{pr}_{\sigma^{-1}t}, \qquad \tau_{\sigma,t} : E_{\sigma^{-1}t} \longrightarrow E_t.$$

The whole tuple is transported; no isomorphism $K_{\mathrm{b}} \simeq K_{\mathrm{g}}$, or independent recombination of tuple components, is used. The fuller source-to-hull argument is recorded in [Full, §§7–8].

6 The comparison

Step (v) of [IUT4, Thm. 1.10] fixes j and a collection $\{v_i\}_{i \in S^{\pm}_{j+1}}$, takes "$i^{\dagger}$" to be $j \in S^{\pm}_{j+1}$ — the *distinguished* slot — and sets

> "λ" to be 0 if $\underline{v}_j \in \mathbb{V}^{\mathrm{good}}$; "$\lambda$" to be "ord(−)" of the element $\underline{\underline{q}}_{\underline{v}_j}^{j^2}$ if $\underline{v}_j \in \mathbb{V}^{\mathrm{bad}}$,

and then computes "the weighted average, over possible collections $\{v_i\}_{i \in S^{\pm}_{j+1}}$, of these estimates". So λ is determined by the distinguished slot alone, and the average runs over tuples. For $t = (\mathrm{g}, \mathrm{b})$ with $j = 1$ the distinguished slot is $t_1 = \mathrm{b}$, whence $\lambda = \mathrm{ord}_{211}(\underline{\underline{q}}_{\mathrm{b}}^{1}) = 29/7$; for $j = 2$ it is $4 \cdot 29/7 = 116/7$. Note the exponent j^2: taking λ to be $\mathrm{ord}(\underline{\underline{q}})$ alone would omit it.

Theorem 5. [P], conditional on [H]. *Assume* (SA). *For* (1) *at* $j = 1$ *and* $t = (\mathrm{g}, \mathrm{b})$*, in the normalization by* $\log 211$*:*

$$\underbrace{-\lambda + d_I + 1 = -\tfrac{29}{7} + \tfrac{104}{105} + 1 = -\tfrac{226}{105}}_{\textit{bound from [IUT4, Prop. 1.2(iii)]}} \qquad \textit{versus} \qquad \underbrace{-\tfrac{106}{105}}_{\textit{hull of } L_t}, \qquad \textit{margin } \tfrac{8}{7} > 0.$$

Remark 5 (where the bound comes from, and why no error terms occur)**.** [IUT4, Prop. 1.1] defines d_i as the order — for ord normalized by $\mathrm{ord}(p) = 1$ — of a generator of the different of R_i over $\mathbb{Z}_p$; for a tamely ramified extension the different is $\mathfrak{p}_i^{e_i - 1}$, so $d_i = 1 - 1/e_i$. [IUT4, Prop. 1.2] defines a_i and states: "if $p > 2$ and $e_i \leq p - 2$, then $a_i = 1/e_i = -b_i$".

The bound used in Theorem 5 is [IUT4, Prop. 1.2(iii)], stated *precisely* under the hypothesis of the present situation: "Suppose that $p > 2$, and that $e_i \leq p - 2$ for all $i \in I$. Then $\phi(p^{\lambda} \cdot (R_I)^{\sim}) \subseteq p^{\lambda - d_I - 1} \cdot (R_I)^{\sim}$." Here $\lambda = 29/7 = 435/105$ also belongs to $e_{i^{\dagger}}^{-1}\mathbb{Z}$, as required for the chosen slot. Every $e_i \leq 105 \leq 209 = p - 2$ here, so it applies directly and the corresponding normalized log-volume bound is $(-\lambda + d_I + 1) \log p$, with *no* error term. The same value follows from [IUT4, Prop. 1.4(iii)] on taking the admissible choice $I^* = \varnothing$, when its error terms $4|I^*|/p$ and

$\sum_{i\in I^*}\{3+\log e_i\}$ both vanish; we note that the symbol I^* of [IUT4, Prop. 1.1] denotes something different. For $t=(\mathrm{g},\mathrm{b})$ we get $d_I = 0+\frac{104}{105}$, $a_I = 1+\frac{1}{105}$, hence $d_I + a_I = 2 = |I|$, attaining with equality the inequality $d_I + a_I \geq |I|$ of [IUT4, Prop. 1.4(iii)].

Remark 6 (the "nonpositive integer power")**.** Enlarging the intermediate container is permitted, but the accompanying loss of volume must then be carried into the subsequent bound. Here $\lfloor\lambda\rfloor - |I| = \lfloor 29/7\rfloor - 2 = 2$. Writing $C(L_t)$ for the terminal fractional-ideal hull, hull formation commutes with multiplication by p^2, so $C(p^2L_t) = p^2C(L_t)$. This is a proper subideal of the nonzero fractional ideal $C(L_t)$. Thus the intermediate container and its hull cannot contain the seed L_t; this conclusion concerns hulls as well as the lattice inclusion.

Remark 7 (what is not refuted)**.** [IUT4, Prop. 1.4] itself is not in question: its estimate for the input regions it is stated for stands. At issue is its application to the fixed target component of the union of possible images.

A variant with the label 0 fixed

One might ask whether fixing label 0 in every stage eliminates the transport. That label is distinct from the slot j designated in Step (v) when $j \geq 1$. The same datum provides a test of this proposed restriction.

Theorem 6. [P], conditional on [H]. *Assume* (SA) *with the indicated whole transposition still admitted. For the same datum* (1)*, take* $j=2$*, source tuple* $(\mathrm{b},\mathrm{b},\mathrm{g})$*, target* $t=(\mathrm{b},\mathrm{g},\mathrm{b})$ *and* $\sigma=(1\,2)$*, which* fixes the label 0. *Here* $\lambda = \mathrm{ord}(\underline{\underline{q}}^{j^2}) = 4\cdot\frac{29}{7} = \frac{116}{7}$*, and in the same normalization*

$$\textit{upper bound } = -\lambda + d_I + 1 = -\tfrac{1427}{105}, \qquad \textit{hull of } L_t = -\tfrac{107}{105}, \qquad \textit{margin } \tfrac{88}{7} > 0.$$

Here $d_I = \frac{208}{105}$, $a_I = \frac{107}{105}$ and $d_I + a_I = 3 = |I|$. In general, writing $s = \mathrm{ord}_p(\underline{\underline{q}})$ for the depth at a bad slot, Step (v) sets $\lambda = j^2 s$ when the distinguished slot is bad. For a tuple with every $e_i \leq p-2$ and an admitted L_t seed, the margin is

$$\Big(-\sum_i e_{t_i}^{-1}\Big) - \big(-\lambda + d_I + 1\big) = j^2 s - (j+2),$$

independently of the ramification indices, since $d_I = (j+1) - \sum_i e_{t_i}^{-1}$ in that range. With $s = 29/7$ this gives $8/7$, $88/7$ and $226/7$ for $j=1,2,3$. Theorems 5 and 6 are the first two cases. Equivalently it is positive when $\mathrm{ord}_p(q) > 2\ell(j+2)/j^2$. For $j=2$ this threshold is 14.

Remark 8. Fixing label 0 alone leaves the transposition $(1\,2)$ available in this example. This does not show that every nontrivial permutation must be forbidden: a nontrivial subgroup fixing the actual slot j can permute the other factors. For this seed mechanism the relevant issue is whether the allowed orbit of j reaches a good, or sufficiently shallow, source factor.

7 What would falsify this

Theorem 1 stands or falls on its arithmetic and the quoted conditions; if a condition of [IUT1, Def. 3.1] fails, we ask that it be identified. The conditional numerical implication is distinct from its proposed source application. The latter may fail at the following points; these are not asserted to exhaust all possible readings.

(F1) The particular whole transposition reaching the good factor is not admitted by (Ind1), contrary to (SA2). The relevant issue is the orbit of the actual distinguished slot, not whether some permutation is nontrivial.

(F2) The admitted region family or its target differs from (SA4). This includes failure of orbit closure for the specified native family, failure of its prime-strip interpretability after transport, a different order of span/hull and comparison, or a different identification of the fixed arithmetic structure and normalized readout with Step (v). That step attributes the treatment of (Ind1), (Ind2) to the arbitrariness of ϕ in [IUT4, Prop. 1.2]. The map there is an automorphism of fixed E_t, whereas the component transport is $E_{\sigma^{-1}t} \to E_t$. A source-compatible construction connecting these types and explaining the applicable region and bound would answer the question; the type distinction alone does not rule it out.

(F3) The specified native representation has no admitted region with the claimed seed or span in the hull, contrary to (SA1). A formal generator or an isomorphic ideal alone would not justify elementwise membership.

(F4) The comparison or subsequent transport does not carry the integral tensor structure and L to those of the fixed target as used above. The log-unit equality invoked here is for $e_i \leq p-2$, not for arbitrary tame extensions.

(F5) The regions, admitted transports, hull and normalized readout of [LANA, §9.2, §10.4] are identified with the fixed component used here, *and* under that identification the effects of the indeterminacies are not detectable at a single place, as [LANA, §10.4] states of the hull computation in [IUT4]. Both halves are needed: the identification is not established here, and without it the two accounts are not yet about the same objects. If both are supplied then (SA) fails and the source application is withdrawn. We regard this as the cross-check most directly checkable against the source, since it concerns where the effect of (Ind1) can appear rather than our arithmetic.

There is a related type distinction in [DH, §6.2, footnote 7]: the actual ind_1 permits permutations between distinct tensor factors of the adelic tensor power, which the authors' example does not exhibit. They then derive an upper bound of their own in [DH, §6.3, Lemma 6.3.1]. We cite this observation for that distinction only, not as verification of (SA) or of the present conclusion.

(F2) specifies which source quantity is compared. We do not identify it with the separate objection answered in [Rpt, §12], or with the argument in [SS]; those comparisons require their own source and normalization analysis. A demonstration that our source application fails at any of these points would settle that application. *We would withdraw the source application of Theorems 5 and 6, and would regard the demonstration as the more valuable outcome.* The conditional rational calculation would remain, with its unsuccessful source-adapter premise identified.

8 What we ask

First, we ask that the admitted initial datum in §3 and its mixed profile be checked against the source conditions. Its construction does not depend on (SA), and dropping its split bad place outside 2ℓ does not preserve the extra hypothesis of IV. The nonconstant and mixed-status tuple statements in §4 are distinct elementary consequences of the section defining $\mathbb{V}$.

Second, we ask whether the full reading (SA1)–(SA4), including the fixed target and its normalized readout, is correct for this datum. Theorems 5 and 6 are offered as conditional local comparisons and a request to decide that reading, not as a settled adjudication of the theory. If the application is wrong, identifying the failed clause and the source-compatible comparison would answer the question.

9 Reproduction

The companion package [Full] separates the checks from the source arguments. The standard-library program `scripts/checks.py` reproduces the quadratic-integer arithmetic, Frobenius traces, rational local constants, and corrected stage margins. Its profile identities do not prove the existence of arbitrary initial data. The additional programs `scripts/check_round3.py` and `scripts/check_normalization.py` record the later exact checks and the separate normalization identities.

The all-selected-order certificate requires more than those short checks: `scripts/check_norm_seventh_power_free.cpp` and its log test all primes up to $\lfloor C^{1/7} \rfloor = 225469788$. The norm identity then translates that integer certificate to the selected prime-ideal orders. This exhaustive sieve does not certify primality of C. The geometric constructions, cited theorems, and (SA) are not verified by these programs. In particular the existence of the lattice and the compatible marked covers is the mathematical construction in §3, expanded in [Full, §4].

The distinction between invariance of individual log-volumes and stability of a family of possible images is also illustrated by a finite model checked by the Lean 4 kernel in the companion package. Even both properties do not bound a union hull by a member hull. This model is not a formalization of the source-adapter hypothesis or an IUT counterexample.

Disclosure and corrections

Large language models were used substantially in locating passages, cross-checking computations, drafting, and adversarial review. The author is responsible for the claims. The exact computations and the source arguments are separately inspectable; (SA) remains a proposed reading of the source, with its failure conditions stated above.

The correction history is material. An earlier version used $29/7$ instead of $j^2(29/7)$ at $j = 2$, giving the erroneous margin $1/7$. The correct upper bound is $-1427/105$ and the margin is $88/7$. The old value is not a second instance of the same datum. Earlier explanations also wrongly used full 2-torsion to exclude a quadratic twist, and used only the singleton bad set $\{v_\pi\}$ despite the extra IV condition. Both have been corrected. The power of 2 in the degree bound is 2^{10}, not 2^{11}.

A conjecture that all four arithmetic exceptional j-invariants were algebraic integers was false for two of them. The proof uses their rationality and the nonrationality of the present j, with the factorization $488095744 = 2^{14}31^3$ checked against [MFHMP, Prop. 2.1]. We have also corrected the conflation of nonconstant tuples with mixed reduction, the generalization of the log-unit equality beyond $e \leq p - 2$, and the claim that fixing label 0 would require banning every nontrivial permutation. We do not claim that no further errors remain, and retain the withdrawal undertaking of §7.

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